PROJECT ASSIGNMENT - 1

Title: **Rock–Paper–Scissors Digital Game Circuit with Game Counter**

Subject: Advanced Digital System Design (CSE 2697)
Type: Assignment + Mini Project
Total Marks: 10
Tools Suggested: Logisim

1. Project Objective

The objective of this assignment-cum-project is to design and implement a **digital logic circuit that simulates the Rock–Paper–Scissors game** between two players using combinational and sequential logic components.

Students will design:

1. **Game Logic Circuit (Part A)** to determine the winner or tie.
2. **Game Counter and Comparator (Part B)** to determine the final winner after a predefined number of wins.

This project helps students understand the practical application of:

- Decoders
- Combinational logic design
- Boolean logic implementation
- Counters
- Digital comparison circuits
- System-level digital circuit design

2. Learning Outcomes

After completing this project, students will be able to:

- Design and implement **combinational logic circuits**
- Use **decoders for signal representation**
- Implement **game logic using Boolean expressions**
- Design **digital counters**
- Use **comparators to detect threshold conditions**
- Extend digital systems to support **additional states**

3. System Description

3.1 Part A – Rock vs Scissors vs Paper Logic

The circuit determines the **winner between two players** based on their chosen hand signals.

Each player selects one of three possible signals:

- Rock
- Scissors
- Paper

These signals are encoded using **two binary inputs**.

Input Assignment

Hand Sign	S1/S3	S2/S4
Rock	0	0
Scissors	0	1
Paper	1	0
Initial State	1	1

Player Inputs:

- **Player P:** S1, S2
- **Player Q:** S3, S4

Circuit Operation

1. Each player input is passed through a **2-to-4 decoder**.
2. The decoder generates individual signals for:
    - Rock
    - Scissors
    - Paper
3. A **selector circuit** compares both player outputs.
4. Depending on the combination:
    - **Player P Wins LED ON**
    - **Player Q Wins LED ON**
    - **Tie LED ON**

Winning Conditions

Player P	Player Q	Result
Rock	Scissors	P Wins
Scissors	Paper	P Wins
Paper	Rock	P Wins
Scissors	Rock	Q Wins
Paper	Scissors	Q Wins
Rock	Paper	Q Wins

Tie occurs when both players choose the same symbol.

4. Part A Circuit Design Tasks

Task 1

Construct the **2-to-4 decoder** for Player P and Player Q.

Draw the decoder circuit.

Student Work Area:

(Insert Decoder Circuit Diagram)

Task 2

Complete the **truth table for the game logic**.

S1	S2	S3	S4	Player P	Player Q	Tie
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	1	0	0			
0	1	1	0			
1	0	0	0			
1	0	0	1			
1	0	1	0			

Task 3

Derive the **Boolean expressions** for:

1. Player P wins
2. Player Q wins
3. Tie condition

Student Solution:

P_win =

Q_win =

Tie =

Task 4

Design the **selector logic circuit** using logic gates.

Student Work Area:

(Insert selector Logic Gate Circuit)

5. Part B – Counter and Comparator

The second part determines the **overall winner of the game**.

System Rules

- Every time a player wins, their **counter increments by 1**.
- The first player to reach **4 wins** becomes the final winner.
- When a player reaches **4**, the corresponding **Winner LED** turns ON.

5.1 Circuit Components

The circuit contains:

- Two **binary counters**
- A **comparator circuit**
- Output LEDs indicating:
  - Player P Champion
  - Player Q Champion

5.2 Counter Logic

Event	Counter Action
Player P wins	Counter P increments
Player Q wins	Counter Q increments
Tie	No counter increment

Task 5
Design a **3-bit counter** for Player P.
Student Work Area:

(Insert Counter Circuit)

Task 6

Design a **3-bit counter** for Player Q.

Student Work Area:

(Insert Counter Circuit)

Task 7

Design a **comparator circuit** that detects when the counter equals **4 (100)**.

Boolean Expression:

Winner = Q2 · Q1' · Q0'

Student Work Area:

(Insert Comparator Circuit)

6. Complete System Integration

Students must integrate:

- Player input decoders
- Selector logic
- Counters
- Comparator
- LEDs

Final Circuit

Student Work Area:

(Insert Full System Circuit Diagram)

7. Simulation Results

Students must include screenshots of simulation results.

Required Simulation Cases

Test Case	Player P	Player Q	Expected Result
1	Rock	Scissors	
2	Paper	Rock	
3	Rock	Paper	
4	Scissors	Scissors	
5	Paper	Paper	

8. Evaluation Rubric

Component	**Marks**
Truth Table Completion	1
Decoder Design	1
Selector Logic Design	2
Counter Implementation	2
Simulation and Results	2
Report Quality	2

9. Instructor Remarks

Instructor Comments:

_____

_____

_____

(Signature of the Faculty)

(Signature of the Student)

Group No.: _____

Name: _____

Date: _____ **Registration No.:** _____

Branch: _____

Section _____